



CSE623 Machine Learning: Theory and Practice

Topic:

Machine Learning Based Risk Parity Portfolio Construction for Indian Equity Markets

Weekly Report 1

Submitted to: Prof. Mehul Raval

Submitted on 14th February 2026

Team Name: RiskNeutral

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1. Overview of Project:

Risk parity has emerged as a widely adopted portfolio construction framework that allocates capital by equalizing risk contributions across assets rather than maximizing expected returns. Popularized in institutional asset management as an “all weather” strategy, risk parity portfolios are designed to deliver stable performance across different market regimes. However, the effectiveness of risk parity critically depends on accurate estimation of asset volatilities and correlations.

Recent advances in machine learning and statistical learning theory provide tools for denoising and dynamically estimating covariance structures. Techniques such as shrinkage estimators, random matrix theory based eigenvalue filtering, and hierarchical clustering provides more robust representations of dependency structures (tail risks) for a multi asset portfolio.

This project builds a risk parity portfolio using machine learning techniques. Using minute by minute Indian equity data from 2017–2020, we construct a long-only, fully invested portfolios. We then evaluate their out of sample performance during the 2020 COVID stress period to assess whether the risk parity “all-weather” hypothesis holds in an emerging market context.

2. Overview of Week 1:

Tasks done till now:

- Among the list of over 155+ stocks, deciding which stocks to pick.
- Final stocks were filtered to ensure our dataset is well-suited to the Marcenko-Pastur theorem for eigenvector decomposition.
- We identified 132 stocks to finally work with.
- An inner join on timestamps was used, and missing values were eliminated.

3. Work Completed:

- We imported and processed 155 equity time series from the Kaggle dataset, ensuring consistent formatting and timestamp alignment across assets.
- We examined listing start dates for each stock to verify data consistency across the sample period.
- Since the data starts on 1st January 2017, all timestamps for each stock should also start at the same time. However, we found discrepancies in the start dates.

- Here is a small snippet of the same.

	Start_Date	End_Date	Rows
Stock			
SUNTV	2017-01-02 09:15:00	2021-01-01 15:29:00	370545
EMAMILTD	2017-01-02 09:15:00	2021-01-01 15:29:00	370528
INFY	2017-01-02 09:15:00	2021-01-01 15:29:00	370546
GRASIM	2017-01-02 09:15:00	2021-01-01 15:29:00	370517
ASIANPAINT	2017-01-02 09:15:00	2021-01-01 15:29:00	370545
EDELWEISS	2017-01-02 09:15:00	2021-01-01 15:29:00	370545
NATCOPHARM	2017-01-02 09:15:00	2021-01-01 15:29:00	370546
HDFC	2017-01-02 09:15:00	2021-01-01 15:29:00	370545
HEROMOTOCO	2017-01-02 09:15:00	2021-01-01 15:29:00	370545
NIFTY_BANK	2017-01-02 09:15:00	2021-01-01 15:29:00	370740
BANKINDIA	2017-01-02 09:15:00	2021-01-01 15:29:00	370545
ABCAPITAL	2017-09-01 09:44:00	2021-01-01 15:29:00	308836
SBILIFE	2017-10-03 09:44:00	2021-01-01 15:29:00	300961
GODREJAGRO	2017-10-16 09:44:00	2021-01-01 15:29:00	297584
NAM_INDIA	2017-11-06 09:44:00	2021-01-01 15:29:00	292641
HDFCLIFE	2017-11-17 09:44:00	2021-01-01 15:29:00	289277
ISEC	2018-04-04 09:44:00	2021-01-01 15:29:00	254772
ADANIGAS	2018-11-05 09:44:00	2021-01-01 15:29:00	200026
NIFTY_COMMODITIES	2018-12-10 09:15:00	2021-01-01 15:29:00	190620
NIFTY_MIDCAP_100	2018-12-10 09:15:00	2021-01-01 15:29:00	175841
DALBHARAT	2019-01-22 09:35:00	2021-01-01 15:29:00	180373
POLYCAB	2019-04-16 09:15:00	2021-01-01 15:29:00	158746
NIFTY_500	2019-07-02 09:15:00	2021-01-01 15:29:00	139995
IRCTC	2019-10-14 09:44:00	2021-01-01 15:29:00	114467

- After inspecting some of the stocks, we found that the main reason was that they were listed on the NSE after 1st January

2017. For example, ADANIGAS and IRCTC were listed in the years 2018 and 2019, respectively.

- As we are going to construct the covariance matrix and later apply the MP theorem, we had to decide on the right method for handling this issue.
- We explored whether the covariance matrix can be calculated when each pair has a different number of observations.
- In the MP theorem, one necessary condition is that all data points come from a data table with a fixed number of rows and columns.

$$\lambda_{\pm} = \sigma^2 \left(1 \pm \sqrt{\frac{n}{T}} \right)^2$$

(Source: Cajas, D. (2025), Advanced Portfolio Optimization, Springer)

- Here, “n” is the number of stocks and “T” is the number of total timestamps we will use.
- After filtering for the stocks for which we have data since 1st January 2017, our next step was to find the null values within the filtered stocks.
- In our analysis, we found that many stocks had null values; however, count of null values was less than 1000. For the context, our dataset has more than 3,70,000 rows.
- Given that null observations were negligible relative to the total sample size (~370,000 timestamps), we removed rows containing missing values to preserve matrix dimensional consistency required for covariance estimation.
- At the end, we performed an inner join on the timestamp column.

4. Challenges encountered:

- Reading all 155+ files on the Kaggle notebook.
- Interpreting the theoretical assumptions of the Marčenko–Pastur theorem, particularly the requirement of a fixed-dimensional data matrix, required careful validation of our dataset structure.
- Inspecting the reason for discrepancies in timestamps.
- Deciding the appropriate way to handle null values given our dataset.

5. Plan for the Week 2:

- Identifying anomalies in the price data and determining the appropriate method to handle them.
- Estimating covariance matrices using historical and shrinkage methods. We aim to use five methods to estimate the covariance matrix. These methods are usually used when working with noisy financial data.
- Understanding hierarchical clustering-based risk parity portfolio (hierarchical risk parity) in detail and preparing a theoretical framework to apply the same.

References:

1. Cajas, D., *Advanced Portfolio Optimization*: Springer (2025)
2. H. K., "Stock Market India Dataset," Kaggle, 2023. [Online]. Available: <https://www.kaggle.com/datasets/hk7797/stock-market-india>